2.5 Proof by Contradiction and the Classical Rules

Every rule so far either builds a statement or takes one apart. The rules for negation are no exception: ¬I (Definition 2.2.1) builds $\neg\varphi$ from a box that opens with φ and ends in $\bot$, and ¬E (Definition 2.1.1) uses $\neg\varphi$ against φ. Yet in three places this chapter has stopped one step short of a goal. This section adds the one rule that takes a negation away, reductio ad absurdum. It then derives from it the classical laws that mathematics uses without noticing, and shows that three of them are one principle in three forms.

2.5.1 Two boxes that look alike

Here are two arguments that a mathematician would both call "proof by contradiction".

The first is the proof of Theorem 2.2.1. It fixes a rational number r, supposes $r^2 = 2$, reaches a contradiction, and concludes that $r^2 \neq 2$. In the language of Definition 2.2.1, it opens a box with the assumption P ("$r^2 = 2$"), reaches $\bot$ inside it, and closes the box to get $\neg P$. That is ¬I.

The second is an argument of a kind that analysis uses constantly.

Example. A number below every positive number

Let x be a real number with $x \geq 0$, and suppose that $x \leq \varepsilon$ for **every** real number $\varepsilon > 0$. Prove that $x = 0$.

Solution. Suppose, for a contradiction, that $x \neq 0$. Since $x \geq 0$ and $x \neq 0$, we have $x > 0$. Then $\varepsilon := x/2$ is a real number with $\varepsilon > 0$, so the hypothesis applies to it and gives $x \leq x/2$. Subtracting $x/2$ from both sides gives $x/2 \leq 0$, and multiplying by 2 gives $x \leq 0$. This contradicts $x > 0$. Hence $x = 0$.

(The step "the hypothesis applies to it" uses a rule for "every", which Chapter 5 makes precise. The facts about $\leq$ are school algebra, which *Analysis on the Real Line*, Chapter 1 derives from axioms.)

Look at its shape. The box opens with the assumption $\neg P$, where P is "$x = 0$", reaches $\bot$, and closes to give P. If we tried to justify the last line by ¬I, we would get the negation of the box's assumption, $\neg\neg P$: "it is not the case that $x \neq 0$". That is not P. The two arguments put their negations in different places. ¬I puts a ¬ **into** the conclusion; the second argument takes one **out**.

2.5.2 Where the rules so far stop short

The second argument is not a curiosity. Three times already in this chapter a derivation has stopped short in the same way: it reached a negation where the goal was not one.

1. Section 2.2 warned against the argument "suppose x is not positive; …; hence x is positive". The box opens with $\neg(x > 0)$ and reaches $\bot$, so ¬I gives $\neg\neg(x > 0)$, and no rule so far gets from there to $x > 0$.

2. In Exercise (C1: Two premises that cannot both fail), the premises $P \to Q$ and $\neg P \to R$ led to $\neg(\neg Q \wedge \neg R)$, "not both Q and R fail", and not to $Q \vee R$.
3. Section 2.4 noted, after Example (Negating both sides of an iff), that its converse, $\neg P \leftrightarrow \neg Q \vdash P \leftrightarrow Q$, resists the rules so far. The natural box on $\neg Q$, inside the box of P, closes to $\neg\neg Q$, not Q.

Try the smallest case, P from $\neg\neg P$. The goal P is a letter, so no introduction rule applies: the goal-directed moves have nothing to say. The premise $\neg\neg P$ is a negation, and a negation can be used only by $\neg$E, against the statement it negates, $\neg P$. So we assume $\neg P$ in a box. The box reaches $\bot$ at once, and $\neg$I closes it with $\neg\neg P$: the premise again. Every attempt goes round this circle.

In fact no derivation of P from $\neg\neg P$ uses only the rules of Chapter 1 and Sections 2.1–2.4. That is a claim about **all** derivations, and no search can prove it. Chapter 4 proves it (Theorem 4.6.4), by reading statements with more than two truth values. For the other two stalls the same is true, but we state it without proof, and nothing in this book depends on it. Here we name the missing step and add it as a rule.

2.5.3 Reductio ad absurdum

To prove φ, it is enough to show that "not φ" leads to a contradiction.

Definition 2.5.1. Reductio ad Absurdum

i		$\neg\varphi$	asm, for RAA
		$\vdots$	
j		$\bot$	
k	φ		RAA i–j

This is the rule RAA (**reductio ad absurdum**, "reduction to the absurd"). The line $\neg\varphi$ is an assumption, which opens the box and is **discharged** when the box closes. The line $\bot$ is a line **of that box itself**, not of a box inside it.

As in every schema (Definition 1.2.1), φ stands for any statement and i, j, k for line numbers. Read the rule clause by clause.

1. **The assumption is the exact negation of the goal.** To prove φ, the box opens with $\neg\varphi$: the negation of the **whole** of φ. If φ is $Q \vee R$, the box opens with $\neg(Q \vee R)$, not with $\neg Q \vee R$.
2. **The box must reach** $\bot$, as a line of the box itself: a $\bot$ inside a smaller box depends on a further assumption, as in $\neg$I.
3. **The result is φ itself**, not $\neg\neg\varphi$, which $\neg$I would give from the same box. This clause is the whole content of the rule.
4. **The box closes**, its assumption is discharged, and the justification cites the box as a range i–j.

In words: the box is a thought experiment in which φ fails. If the experiment ends in $\bot$, then φ cannot fail, and RAA lets us write φ. Set the new rule beside $\neg$I:

Rule	The box opens with	The box ends with	After the box
¬I	φ	$\bot$	$\neg\varphi$
RAA	$\neg\varphi$	$\bot$	φ

The rows differ only in where the ¬ sits, and that small difference makes RAA different in kind. Every other primitive rule is the introduction or the elimination rule of one connective ($\bot$E counts as the elimination rule of $\bot$), and repetition is a clause of Definition 1.4.2, not a rule for a connective. RAA belongs to no connective. It applies to every statement φ, whatever its main connective, and even to a letter, which has none.

For one kind of goal the new rule is never needed at the last step. Suppose the goal is itself a negation $\neg\psi$. An RAA box for it opens with $\neg\neg\psi$. Replace it by a ¬I box that opens with ψ; inside it, derive $\neg\neg\psi$ from ψ by the three lines of Example (Two negations from one premise) (a box on $\neg\psi$ that clashes with ψ); then continue as the RAA box did, citing that line wherever the RAA box cited its assumption. The box still ends in $\bot$, and ¬I closes it with $\neg\psi$. So the **last** step of a derivation of a negation never needs RAA, though steps inside its box still may. RAA earns its place on the other goals.

Warning. The assumption that opens an RAA box must be the **exact** negation of the goal. To prove $x \leq 1$ by contradiction, assume $x > 1$, since for real numbers "not $x \leq 1$" says $x > 1$. Assuming $x \geq 1$ instead is a common slip. It is not the negation: $x = 1$ satisfies both $x \leq 1$ and $x \geq 1$. A box opened with $x \geq 1$ cannot honestly reach $\bot$ while $x = 1$ is possible, so a "contradiction" that seems to be reached hides an error.

Quick check. A box opens with the assumption $\neg(x < 0)$ and reaches $\bot$, as a line of the box itself. Which of the following may be written after it closes, and by which rule? (i) $x < 0$; (ii) $\neg\neg(x < 0)$; (iii) $x \geq 0$.

Solution.

(i) By RAA: the box opened with the negation of $x < 0$. (ii) By ¬I: ¬I writes the negation of the box's assumption, and the assumption is $\neg(x < 0)$. (iii) By neither. For real numbers $x \geq 0$ says the same as $\neg(x < 0)$, which is exactly the statement the box has refuted.

2.5.4 The move: when stuck, assume the negation of the goal

Chapter 1 and Sections 2.1–2.4 built two lists of moves. From the goal: its main connective chooses an introduction rule and so a box or two subgoals. From the premises: each available line's main connective chooses the elimination rule that uses it, and a goal $\bot$ asks for a clash. The new rule adds one move, for the moment when both lists are used up.

We are **stuck** at a gap when the goal-directed moves give nothing (the goal is a letter, or a disjunction with neither side in hand) and no premise-directed move brings the goal closer.

Then open an RAA box on the negation of the goal. The new goal is $\bot$, so we hunt for a clash, and the box has given us one more line to clash with: $\neg\varphi$ itself.

When stuck, assume the negation of the goal, and hunt for a clash.

The move is a last resort, not a first one. Section 3.1 explains why it never does harm and still comes last. Four of the five derivations below reach the moment of being stuck, and their Ideas say where; the first shows the rule at its smallest.

2.5.5 Five derivations with the new rule

The degenerate case. The smallest use of RAA proves P from P.

Idea. There is nothing to find: the premise is the goal. The derivation is here only to show the rule at its smallest. The box opens with $\neg P$, and the one line it needs, $\bot$, comes from the clash of the assumption with the premise.

1	P	
2	$\quad\neg P$	asm, for RAA
3	$\quad\bot$	¬E 1, 2
4	P	RAA 2–3

It is legal and pointless, since line 1 already is the goal. It is the purest case of a pattern that turns up in real proofs: a contradiction box whose assumption does nothing except clash with a line we already had. The warnings at the end of this section return to it.

The stall, resolved. The next result is the one on which we got stuck.

Proposition 2.5.1. Double Negation

For every statement P:

(a) $P \vdash \neg\neg P$, by a derivation that does not use RAA;
(b) $\neg\neg P \vdash P$, by a derivation that uses RAA.

Idea. In (a) the goal $\neg\neg P$ is the negation of $\neg P$, so the goal-directed move of Section 2.2 opens a ¬I box with $\neg P$, and the premise clashes with it at once.

In (b) the goal P is a letter, so no introduction rule applies. The only premise is a negation, which can only clash; the attempt above showed that clashing it inside a ¬I box leads back to the premise. We are stuck, so we assume the negation of the goal: an RAA box on $\neg P$. The premise clashes with it at once.

Proof.

(a) This is the derivation of Example (Two negations from one premise) (a), which uses the rules of Definition 2.2.1 alone; we repeat it to set it beside (b).

1	P	
2	$\quad \neg P$	asm, for $\neg$I
3	$\quad \bot$	$\neg$E 1, 2
4	$\neg\neg P$	$\neg$I 2–3

(b) An RAA box on $\neg P$:

1	$\neg\neg P$	
2	$\quad \neg P$	asm, for RAA
3	$\quad \bot$	$\neg$E 2, 1
4	P	RAA 2–3

Line 3 applies $\neg$E to the statement $\neg P$ on line 2 and to its negation $\neg\neg P$ on line 1. This proves both parts.

■

In both derivations, lines 2 and 3 form a box that opens with $\neg P$ and reaches $\bot$ by a clash with line 1. Only the rule that closes the box differs, and with it the last line. So the two directions are not mirror images: one needs only the rules for $\neg$, and the derivation of the other uses the new rule.

Nonexample by minimal change. Change only the last line of (a), from $\neg\neg P$ to P, keeping the justification $\neg$I 2–3. Lines 1–3 still check. What fails is the conclusion of $\neg$I, which must be the negation of the box's assumption, here $\neg\neg P$. To write P, the justification must be RAA.

From now on both parts are derived rules in the sense of Definition 2.3.1. A line $\neg\neg\varphi$ may be justified by "$\neg\neg$I i", citing a line φ, and a line φ by "$\neg\neg$E i", citing a line $\neg\neg\varphi$. The first stall of the list above is now one line: from $\neg\neg(x > 0)$, $\neg\neg$E gives $x > 0$.

Contraposition, backwards. Proposition 2.3.2 derives $\neg Q \to \neg P$ from $P \to Q$ with the rules for $\to$ and $\neg$ alone. In practice we use it the other way round. To prove "if n^2 is even then n is even", one may prove "if n is not even then n^2 is not even", and the step from that back to the original goes from $\neg Q \to \neg P$ to $P \to Q$.

Proposition 2.5.2. Converse of the Contrapositive

For all statements P and Q, $\neg Q \to \neg P \vdash P \to Q$.

Idea. The goal is an implication, so the goal-directed move opens a box with P, and the new goal is Q, a letter. The premise needs $\neg Q$ before it gives anything, and $\neg Q$ is nowhere; the box's assumption P is a letter and cannot be taken apart. We are stuck at Q, so we assume its negation, $\neg Q$. That is exactly what the premise wanted: $\to$E gives $\neg P$, which clashes with the P of the outer box, and RAA closes the inner box with Q.

Proof. The derivation is:

1	$\neg Q \to \neg P$	
2	$\quad P$	asm, for $\to$I
3	$\quad\quad \neg Q$	asm, for RAA
4	$\quad\quad \neg P$	$\to$E 1, 3
5	$\quad\quad \bot$	$\neg$E 2, 4
6	$\quad Q$	RAA 3–5
7	$P \to Q$	$\to$I 2–6

Line 5 may cite line 2 because line 2 lies in a box that is still open. RAA enters at line 6. This proves the proposition.

■

So contraposition behaves like double negation: going from $P \to Q$ to $\neg Q \to \neg P$ needs no RAA, and our derivation of the way back uses one RAA. That no derivation of the way back can avoid it is true, but we state it without proof. When Q is itself a negation, the inner goal is a negation, and by the paragraph after Definition 2.5.1 that RAA can be replaced by $\neg$I.

Excluded middle. Every integer is even or odd. Every real number is ≥ 0 or < 0. The purely logical form of such facts is the law $P \vee \neg P$, which the next result derives from no premises at all.

Proposition 2.5.3. Excluded Middle

For every statement P, $\vdash P \vee \neg P$.

Idea. The goal is a disjunction, so Section 1.5's move postpones it: choose a side only when that side is in hand. But there are no premises, so no premise-directed move applies, and neither side can be derived on its own (informally: P may be true or false; Chapter 4 makes this exact). We are stuck, so we assume the negation of the whole goal, $\neg(P \vee \neg P)$, which says that neither side holds, and hunt for a clash. The new line can clash only with $P \vee \neg P$, which $\vee$I builds from either side. So we manufacture a side, as in Example (Manufacturing the other half of a clash). If P held, then $P \vee \neg P$ would hold, against the assumption; so $\neg P$, by $\neg$I. But then $P \vee \neg P$ holds after all, and that is the final clash.

Proof. The derivation is:

1	$\neg(P \vee \neg P)$	asm, for RAA
2	$\quad P$	asm, for $\neg$I
3	$\quad P \vee \neg P$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg P$	$\neg$I 2–4
6	$P \vee \neg P$	$\vee$I 5
7	$\bot$	$\neg$E 6, 1
8	$P \vee \neg P$	RAA 1–7

The derivation has no premises, so it shows $\vdash P \vee \neg P$. RAA is used once, at line 8. This proves the proposition.

■

The assumption of line 1 is used twice, at lines 4 and 7. The first use refutes P; the second uses that refutation. Excluded middle now becomes a derived rule with no premises: a line $\varphi \vee \neg\varphi$ may be written anywhere, for any statement φ, justified by LEM with no line cited. Pasting in the eight lines above would replace it.

Its use is **proof by cases on a statement**. To derive χ, derive it once in a box that assumes φ and once in a box that assumes $\neg\varphi$, and combine the two boxes with the line $\varphi \vee \neg\varphi$ by $\vee$E. The paragraph after De Morgan's laws below uses it that way, and so does part (b) of the result after them.

A law with no negation. The last of the five is a law in which $\neg$ does not occur at all.

Proposition 2.5.4. Peirce's Law

For all statements P and Q, $\vdash ((P \to Q) \to P) \to P$.

Try it first (ten minutes): use the goal-directed and premise-directed moves of the earlier sections on this goal until they stall. At which gap do they stall, and which line, if it were available there, would let you finish?

Idea. The goal is an implication, so we open a box on $(P \to Q) \to P$ with the goal P, a letter. The only line that can produce P is the assumption itself, by $\to$E, once we have its hypothesis $P \to Q$. So $P \to Q$ becomes a subgoal, and the goal-directed move opens a box on P with the goal Q. There the moves stall: the resources are $(P \to Q) \to P$ and P, and neither says anything about Q. What the stalled gap lacks is a clash, which $\bot$E would turn into Q.

We are stuck, so the new move applies, but where? Assuming $\neg Q$ at the stalled gap gives nothing new: a clash with $\neg Q$ needs Q, the stalled goal itself. So apply the move one level out, to the goal P, and assume $\neg P$. That is exactly what the stalled gap lacked: inside the box of P, the pair P, $\neg P$ gives $\bot$, and $\bot$E gives Q. Then $P \to Q$, then P by $\to$E, which clashes with $\neg P$.

Proof. The derivation is:

1	$(P \to Q) \to P$	asm, for $\to$I
2	$\quad \neg P$	asm, for RAA
3	$\quad\quad P$	asm, for $\to$I
4	$\quad\quad \bot$	$\neg$E 3, 2
5	$\quad\quad Q$	$\bot$E 4
6	$\quad P \to Q$	$\to$I 3–5
7	$\quad P$	$\to$E 1, 6
8	$\quad \bot$	$\neg$E 7, 2
9	P	RAA 2–8
10	$((P \to Q) \to P) \to P$	$\to$I 1–9

The derivation has no premises, and RAA enters at line 9. This proves the proposition. ■

The assumption $\neg P$ of line 2 is used twice: at line 4 it makes the innermost box contradictory, and at line 8 it clashes with the P that line 7 obtained. Notice where the new box went: around the stalled box, not inside it. An assumption made one level out is available in every gap inside its box, and so it could give the stalled gap the line it lacked. Peirce's law contains no $\neg$, yet our derivation uses RAA: the classical rule is about more than negation. (That every derivation of it must use RAA, or one of the two rules this section shows equivalent to it, is true, and Chapter 4 proves it in the remark after Theorem 4.6.4, by the same reading with more than two truth values that settles $\neg\neg P \vdash P$.)

2.5.6 De Morgan's laws

The negation of an "and" is an "or" of negations, and the negation of an "or" is an "and" of negations. For example, a real number x fails to lie in the interval $[0, 1]$, that is, $\neg(0 \leq x \wedge x \leq 1)$, exactly when $x < 0$ or $x > 1$. Each law has two directions, so there are four derivations to give.

Proposition 2.5.5. De Morgan's Laws

For all statements P and Q:

(a) $\neg(P \vee Q) \vdash \neg P \wedge \neg Q$;
(b) $\neg P \wedge \neg Q \vdash \neg(P \vee Q)$;
(c) $\neg P \vee \neg Q \vdash \neg(P \wedge Q)$;
(d) $\neg(P \wedge Q) \vdash \neg P \vee \neg Q$.

Try it first (ten minutes): look for a derivation of (d) with the rules of the earlier sections. Which rule would give its last line, and what would that rule need?

Idea. Parts (a)–(c) yield to their goals. In (a) the goal is a conjunction of negations: two ¬I boxes, one per conjunct. In (b) and (c) the goal is a negation: one ¬I box, with the disjunction used inside it by ∨E.

Part (d) resists. Its goal is a disjunction, so the natural last line is ∨I, from $\neg P$ or from $\neg Q$. But neither follows from the premise alone: with P true and Q false the premise is true and $\neg P$ false; with P false and Q true the premise is true and $\neg Q$ false; and the rules never lead from true premises to a false conclusion (Chapter 4 proves this). The premise itself can be used only by ¬E, against $P \wedge Q$, which we do not have. We are stuck at the whole goal, so we assume its negation, $\neg(\neg P \vee \neg Q)$, which says that neither $\neg P$ nor $\neg Q$ holds. It should give P and Q. Each is a letter, so each gets a small RAA box of its own, whose clash is manufactured by ∨I. Together P and Q clash with the premise.

Proof.

(a) Two ¬I boxes, one for each conjunct, then ∧I:

1	$\neg(P \vee Q)$	
2	$\quad P$	asm, for ¬I
3	$\quad P \vee Q$	∨I 2
4	$\quad \bot$	¬E 3, 1
5	$\neg P$	¬I 2–4
6	$\quad Q$	asm, for ¬I
7	$\quad P \vee Q$	∨I 6
8	$\quad \bot$	¬E 7, 1
9	$\neg Q$	¬I 6–8
10	$\neg P \wedge \neg Q$	∧I 5, 9

(b) One ¬I box that opens with $P \vee Q$. Inside it, ∨E splits into two cases, and each case clashes with one half of the premise:

1	$\neg P \wedge \neg Q$	
2	$\neg P$	∧E 1
3	$\neg Q$	∧E 1
4	$\quad P \vee Q$	asm, for ¬I
5	$\quad\quad P$	asm, for ∨E
6	$\quad\quad \bot$	¬E 5, 2
7	$\quad\quad Q$	asm, for ∨E
8	$\quad\quad \bot$	¬E 7, 3
9	$\quad \bot$	∨E 4, 5–6, 7–8
10	$\neg(P \vee Q)$	¬I 4–9

(c) The same shape as (b), with the roles of $\wedge$ and $\vee$ exchanged:

1	$\neg P \vee \neg Q$	
2	$\quad P \wedge Q$	asm, for $\neg$I
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
5	$\quad\quad \neg P$	asm, for $\vee$E
6	$\quad\quad \bot$	$\neg$E 3, 5
7	$\quad\quad \neg Q$	asm, for $\vee$E
8	$\quad\quad \bot$	$\neg$E 4, 7
9	$\quad \bot$	$\vee$E 1, 5–6, 7–8
10	$\neg(P \wedge Q)$	$\neg$I 2–9

(d) RAA on the goal, and inside it two smaller RAA boxes:

1	$\neg(P \wedge Q)$	
2	$\quad \neg(\neg P \vee \neg Q)$	asm, for RAA
3	$\quad\quad \neg P$	asm, for RAA
4	$\quad\quad \neg P \vee \neg Q$	$\vee$I 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad P$	RAA 3–5
7	$\quad\quad \neg Q$	asm, for RAA
8	$\quad\quad \neg P \vee \neg Q$	$\vee$I 7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad Q$	RAA 7–9
11	$\quad P \wedge Q$	$\wedge$I 6, 10
12	$\quad \bot$	$\neg$E 11, 1
13	$\neg P \vee \neg Q$	RAA 2–12

The derivations of (a), (b), and (c) close every box by $\neg$I or $\vee$E and use no RAA. The derivation of (d) uses RAA at lines 6, 10, and 13. This proves all four laws.

■

Laws (a)–(c) conclude negations or a conjunction of negations, and $\neg$I builds negations. Law (d) concludes a **disjunction**, and $\vee$I needs to know **which** side holds, while the premise says only that P and Q do not both hold. RAA gets round the missing decision without making it.

Excluded middle makes the decision instead. Split on $P \vee \neg P$. In the case $\neg P$, $\vee$I gives $\neg P \vee \neg Q$ at once. In the case P, a $\neg$I box on Q builds $P \wedge Q$, which clashes with the premise, giving $\neg Q$; then $\vee$I gives $\neg P \vee \neg Q$. Written out, this route is 11 lines, with LEM as its one classical line.

Both routes use a classical rule. That no derivation of (d) avoids one is true, but we state it without proof, and nothing in this book depends on it.

Quick check. Suppose RAA were deleted from the list of rules. Which of the four derivations in the proof of Proposition 2.5.5 would still be correct, and what is the first line of the others that would lose its justification?

Solution. The derivations of (a), (b), and (c) would still be correct: they use ¬I, ¬E, ∨I, ∨E, ∧I, and ∧E only. In the derivation of (d), line 6 is the first to lose its justification: it closes the box that opened with $\neg P$ and writes P, which is RAA. (Lines 10 and 13 lose theirs too.)

2.5.7 Implication as a disjunction

In everyday mathematics "if P then Q" and "either P fails or Q holds" are used interchangeably. One direction of that exchange needs only the earlier rules; our derivation of the other uses a classical rule.

Proposition 2.5.6. Implication as a Disjunction

For all statements P and Q:

(a) $\neg P \vee Q \vdash P \to Q$, by a derivation that does not use RAA;

(b) $P \to Q \vdash \neg P \vee Q$, by a derivation that uses LEM.

Idea.

(a) The premise is a disjunction, so we use it by cases, each ending in the current goal $P \to Q$. In each case that goal is an implication, so we open a box with P. In the case $\neg P$, the box clashes at once and ⊥E gives Q: this is the derivation of Proposition 2.1.2. In the case Q, the box repeats Q.

(b) The goal is a disjunction, so we postpone it. The premise is an implication, which gives Q only once we have P, and we do not. So we are stuck, and this time excluded middle is the better tool than an RAA box: we decide P first. In the case P the premise gives Q; in the case $\neg P$ the left side is in hand.

Proof.

(a) ∨E on the premise, with an implication built in each case:

1	$\neg P \lor Q$	
2	$\quad \neg P$	asm, for $\lor$E
3	$\quad\quad P$	asm, for $\to$I
4	$\quad\quad \bot$	$\neg$E 3, 2
5	$\quad\quad Q$	$\bot$E 4
6	$\quad P \to Q$	$\to$I 3–5
7	$\quad Q$	asm, for $\lor$E
8	$\quad\quad P$	asm, for $\to$I
9	$\quad\quad Q$	R 7
10	$\quad P \to Q$	$\to$I 8–9
11	$P \to Q$	$\lor$E 1, 2–6, 7–10

Line 9 repeats line 7 inside the box of line 8, because $\to$I needs Q as a line of that box itself.

(b) Cases on P:

1	$P \to Q$	
2	$P \lor \neg P$	LEM
3	$\quad P$	asm, for $\lor$E
4	$\quad Q$	$\to$E 1, 3
5	$\quad \neg P \lor Q$	$\lor$I 4
6	$\quad \neg P$	asm, for $\lor$E
7	$\quad \neg P \lor Q$	$\lor$I 6
8	$\neg P \lor Q$	$\lor$E 2, 3–5, 6–7

Line 2 is Proposition 2.5.3 used as a derived rule, and the only RAA is inside it. This proves both parts.

■

The two routes of the section now stand side by side. RAA assumes the goal fails and hunts for a clash; LEM splits on a statement of our choosing and gives each case a resource it lacked. When a premise is waiting for a statement S (here $P \to Q$ waits for P), splitting on S is usually the shorter route.

2.5.8 One principle in three forms

RAA gave us $\neg\neg$E and excluded middle. Had we added one of those two as the new primitive rule instead of RAA, would we have proved less?

To say this exactly, we need two rule forms and one notion. $\neg\neg$E **taken as a rule** says: if $\neg\neg\varphi$ stands on an available line, we may write φ. LEM **taken as a rule** says: we may write $\varphi \lor \neg\varphi$

on any line, for any statement φ, citing nothing. Given a list of rules, a rule X **derives** a rule Y over that list if every use of Y in a derivation can be replaced by a fragment that uses only X and the rules of the list, without changing the premises or the last line. Then every sequent derivable with Y added to the list is derivable with X added instead.

Theorem 2.5.7. The Classical Rules Are Interderivable

Over the primitive rules other than RAA, each of the three rules RAA, ¬¬E, and LEM derives the other two. In particular, adding any one of the three to the other primitive rules gives the same derivable sequents.

Idea. Replacements can be chained, so it is enough to go once round a cycle. ① RAA derives the other two: that is Proposition 2.5.3 and Proposition 2.5.1 (b). ② ¬¬E derives RAA: ¬I closes an RAA box, which runs from $\neg\varphi$ to $\bot$, with $\neg\neg\varphi$, and ¬¬E finishes. ③ LEM derives ¬¬E: split on $\varphi \vee \neg\varphi$; the case φ is the goal, and the case $\neg\varphi$ clashes with $\neg\neg\varphi$.

Before reading on: in the fragment for ③ below, which rule does line $k+4$ use, and why may line $k+3$ cite line m?

Proof. ① Suppose a derivation uses ¬¬E at line k, writing φ and citing a line m that holds $\neg\neg\varphi$. Replace line k by the last three lines of the derivation in Proposition 2.5.1 (b), with its premise replaced by line m: a box opening with $\neg\varphi$, the line $\bot$ by ¬E citing the assumption and line m, and then φ by RAA. A use of LEM, writing $\varphi \vee \neg\varphi$, is replaced by the eight lines of the derivation in Proposition 2.5.3, with φ for P; that derivation has no premises. So RAA derives ¬¬E and LEM.

② Suppose a derivation uses RAA at line k, closing a box, lines i to j, that opens with $\neg\varphi$ and ends with $\bot$. Keep the box and replace line k by two lines:

i	$\neg\varphi$	asm, for ¬I
⋮	⋮	
j	$\bot$	
k	$\neg\neg\varphi$	¬I i–j
$k+1$	φ	¬¬E k

Line k is correct by ¬I, since the box opens with $\neg\varphi$ and $\neg\neg\varphi$ is the negation of that assumption. Both rules ask for $\bot$ as a line of the box itself, so the box needs no change. So ¬¬E derives RAA.

③ Suppose a derivation uses ¬¬E at line k, writing φ and citing a line m that holds $\neg\neg\varphi$. Replace line k by six lines:

m	$\neg\neg\varphi$	
	$\vdots$	
k	$\varphi \vee \neg\varphi$	LEM
$k+1$	$\quad\varphi$	asm, for $\vee$E
$k+2$	$\quad\neg\varphi$	asm, for $\vee$E
$k+3$	$\quad\bot$	$\neg$E $k+2$, m
$k+4$	$\quad\varphi$	$\bot$E $k+3$
$k+5$	φ	$\vee$E k, $k+1$–$k+1$, $k+2$–$k+4$

The first case is a one-line box: it opens with φ and so already ends with the goal. In the second case, line $k+3$ may cite line m, because line m was available at the old line k and every new line sits in the same boxes as the old line k, or in boxes opened after it. No rule other than LEM and the rules of the list is used. So LEM derives $\neg\neg$E.

In each replacement the later lines are renumbered, and a later citation of the old line k now cites the fragment's last line, which holds the same statement in the same boxes. The premises and the last line of the derivation are unchanged.

Finally, replacements can be chained. LEM derives RAA: replace each use of RAA by the fragment of ②, and then each use of $\neg\neg$E so created by the fragment of ③. $\neg\neg$E derives LEM: replace each use of LEM by the derivation of ①, and then each use of RAA in it by the fragment of ②. With ①, ②, and ③, this shows that each of the three rules derives the other two, as claimed.

■

So the classical part of our rules is **one** extra principle, whichever of its three forms we write down. Chapter 8 tells the story of the mathematicians who declined it, and shows what their mathematics looks like.

The base list matters. Part ③ uses $\bot$E at line $k+4$, to close the impossible case as in Example (An impossible case), and without $\bot$E among the base rules LEM no longer derives $\neg\neg$E. We state this without proof; nothing depends on it.

Remark. Some books call exactly our RAA "proof by contradiction" or PBC, and some take $\neg\neg$E as their primitive rule and derive RAA from it. By Theorem 2.5.7 nothing changes: all these choices prove the same sequents.

Remark. $\bot$E could have been dropped from the primitive rules, since RAA and repetition derive it (Exercise B3 below asks for the derivation). We keep it so that the primitive rules other than RAA form a system of their own, the one Chapter 8 studies in its last section.

2.5.9 How reductio looks in a mathematical proof

Return to Example (A number below every positive number), and read its solution one sentence at a time, with the rule of each step in brackets.

1. "Suppose, for a contradiction, that $x \neq 0$." [An RAA box opens, with $\neg(x = 0)$, the exact negation of the goal $x = 0$.]
2. "Since $x \geq 0$ and $x \neq 0$, we have $x > 0$." Proposition 2.3.3
3. "Then $\varepsilon := x/2$ is a real number with $\varepsilon > 0$." [School algebra.]
4. "So the hypothesis applies to it and gives $x \leq x/2$." [The hypothesis is about **every** $\varepsilon > 0$; applying it to one is a rule for "every", from Chapter 5.]
5. "... gives $x/2 \leq 0$, and ... $x \leq 0$." [School algebra.]
6. "This contradicts $x > 0$." [¬E: for real numbers $x > 0$ is $\neg(x \leq 0)$, the negation of the line $x \leq 0$.]
7. "Hence $x = 0$." [RAA closes the box.]

The words "for a contradiction" do not by themselves tell ¬I from RAA. Look at what is assumed and what is concluded. Here the box assumes a negation, $x \neq 0$, and the conclusion is the positive statement $x = 0$, so the step is RAA. The proof of Theorem 2.2.1 assumes the positive statement $r^2 = 2$ and concludes its negation, so it is ¬I. *Analysis on the Real Line*, Chapter 1 opens its proof that the natural numbers are not bounded above by supposing, for contradiction, that they are bounded above, and concludes that they are not: that is ¬I too, despite the phrase.

Warning. An RAA box on $\neg\varphi$ whose assumption is used **only at its last step**, against a φ derived without it, is legal, but it is a direct proof in disguise: the lines before the clash already derive φ, so the box can be deleted. The degenerate derivation of $P \vdash P$ above is the smallest case. Chapter 8 shows how to spot this costume in written proofs and take it off. (A box that reaches $\bot$ without using its assumption at all is a different matter: the same lines, written before the box, derive $\bot$ without it, and $\bot$E then gives the goal at once.)

Warning. Not every "proof by contradiction" is RAA. Calling the proof of Theorem 2.2.1 an RAA is a mislabeling: its box opens with $r^2 = 2$, not with a negation, and it concludes $r^2 \neq 2$, which ¬I writes. The difference matters when you count classical steps: the box of that proof closes by ¬I and contributes none.

2.5.10 What the rules sound like

Written mathematics carries these rules in a handful of phrases. The Lean column is for readers who use that proof assistant; nothing later depends on it, and the full dictionary is in Chapter 8.

Rule	You write	Lean
RAA	"Suppose, for a contradiction, that φ fails. ... This is impossible. Hence φ."	`by_contra h`
¬¬E	"It cannot be that φ fails; so φ."	`not_not.mp h`

Rule	You write	Lean
LEM	"Either φ or not. If φ, ... If not, ... In either case, ..."	`by_cases h : P`

2.5.11 Exercises

2.5.11.1 A. Check your understanding

Exercise. A1

(a) "Suppose 7 is even. Then $7 = 2k$ for some integer k, so $k = 7/2$, which is not an integer. This is impossible. Hence 7 is not even." Is the last step ¬I, RAA, or neither? What did the box open with?

(b) "Let x be real. Suppose $x^2 \geq 2x - 1$ fails, that is, $x^2 < 2x - 1$. Then $(x-1)^2 < 0$, which is impossible. Hence $x^2 \geq 2x - 1$." Same questions.

(c) "Assume P. Then, as shown above, Q. Hence $P \to Q$." Same questions.

(d) State RAA and ¬I side by side, and say which of the two the derivation of Proposition 2.5.1 (a) uses.

(e) In the derivation of Proposition 2.5.5 (d), why can the last line not be ∨I applied to a line derived from the premise alone?

(f) In the same derivation, the assumption $\neg(\neg P \vee \neg Q)$ on line 2 is cited twice. Name the two lines that cite it, and the rule each uses.

(g) In the fragment for ② in the proof of Theorem 2.5.7, which rule justifies line k, and why does the box of lines i to j need no change?

Solution.

(a) ¬I: the box opened with "7 is even", and the conclusion is its negation. ($7/2$ is not an integer because $3 < 7/2 < 4$.)

(b) RAA: the box opened with the negation of $x^2 \geq 2x-1$, and the conclusion is $x^2 \geq 2x-1$ itself. (The clash: $x^2 < 2x-1$ rearranges to $(x-1)^2 < 0$, and squares are nonnegative.)

(c) Neither: this is →I. The box opened with P and ended with Q; no $\bot$ appears.

(d) ¬I: a box that opens with φ and reaches $\bot$ gives $\neg\varphi$. RAA: a box that opens with $\neg\varphi$ and reaches $\bot$ gives φ. The derivation of Proposition 2.5.1 (a) uses ¬I, with $\neg P$ as the assumption, so its conclusion is the negation of $\neg P$, namely $\neg\neg P$.

(e) ∨I would need $\neg P$ or $\neg Q$ on a line derived from the premise alone. Neither can be: with P true and Q false the premise is true and $\neg P$ false, with P false and Q true the premise is true and $\neg Q$ false, and the rules never lead from true premises to a false conclusion (proved in Chapter 4, Theorem 4.6.2).

(f) Line 5 cites it, by ¬E against line 4, $\neg P \vee \neg Q$; and line 9 cites it, by ¬E against line 8, the same statement. Both lines lie in boxes inside the box of line 2, which is still open, so the citations are legal.

(g) ¬I: the box opens with $\neg\varphi$ and ends with $\bot$, so ¬I gives the negation of its assumption, $\neg\neg\varphi$. The box needs no change because ¬I, like RAA, asks only for a box that ends

with $\bot$ as a line of the box itself; the two rules differ in what they write after the box, not in the box.

2.5.11.2 B. Practice

Exercise. B1: With and without the new rule

(a) Derive $\neg\neg P,\ \neg\neg(P \to Q) \vdash \neg\neg Q$ **without** RAA. Then say in one sentence why your derivation does not give Q.

(b) Derive $\neg P \to Q,\ \neg P \to \neg Q \vdash P$, using RAA.

Hint: in (a) the goal is a negation; in (b) it is a letter.

Solution.

(a) The goal $\neg\neg Q$ is a negation, so we open a $\neg$I box on $\neg Q$, with the goal $\bot$. The premises are negations, so each can only clash: $\neg\neg P$ against $\neg P$, and $\neg\neg(P \to Q)$ against $\neg(P \to Q)$. Both of those are negations, so each is itself a subgoal for a $\neg$I box. Make $\neg P$ the subgoal: a box on P. Inside it, make $\neg(P \to Q)$ the subgoal: a box on $P \to Q$, where $\to$E gives Q, which clashes with $\neg Q$. The boxes then close from the inside out.

1	$\neg\neg P$	
2	$\neg\neg(P \to Q)$	
3	$\quad\neg Q$	asm, for $\neg$I
4	$\quad\quad P$	asm, for $\neg$I
5	$\quad\quad\quad P \to Q$	asm, for $\neg$I
6	$\quad\quad\quad Q$	$\to$E 5, 4
7	$\quad\quad\quad \bot$	$\neg$E 6, 3
8	$\quad\quad \neg(P \to Q)$	$\neg$I 5–7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad \neg P$	$\neg$I 4–9
11	$\quad \bot$	$\neg$E 10, 1
12	$\neg\neg Q$	$\neg$I 3–11

All three boxes close by $\neg$I, so no RAA is used. The derivation does not give Q because its last box opens with $\neg Q$, and $\neg$I can only write the negation of that assumption, $\neg\neg Q$; removing the two negations is $\neg\neg$E, a classical step.

(b) The goal P is a letter, and both premises want $\neg P$, which is nowhere. We are stuck, so we open an RAA box with $\neg P$, and it feeds both premises:

1	$\neg P \to Q$	
2	$\neg P \to \neg Q$	
3	$\quad \neg P$	asm, for RAA
4	$\quad Q$	$\to$E 1, 3
5	$\quad \neg Q$	$\to$E 2, 3
6	$\quad \bot$	$\neg$E 4, 5
7	P	RAA 3–6

In words: "Suppose, for a contradiction, that P fails. Then the premises give both Q and $\neg Q$, which is impossible. Hence P."

Exercise. B2: Back from negated sides

Derive $\neg P \leftrightarrow \neg Q \vdash P \leftrightarrow Q$. Say where RAA enters and why.

Solution. The goal is a biconditional, so by Definition 2.4.1 we need two boxes, one from P to Q and one from Q to P. In the first, the goal Q is a letter and the premise speaks only about $\neg Q$, so we are stuck: we assume $\neg Q$ and argue by RAA. The second box is symmetric.

1	$\neg P \leftrightarrow \neg Q$	
2	$\quad P$	asm, for $\leftrightarrow$I
3	$\quad\quad \neg Q$	asm, for RAA
4	$\quad\quad \neg P$	$\leftrightarrow$E 1, 3
5	$\quad\quad \bot$	$\neg$E 2, 4
6	$\quad Q$	RAA 3–5
7	$\quad Q$	asm, for $\leftrightarrow$I
8	$\quad\quad \neg P$	asm, for RAA
9	$\quad\quad \neg Q$	$\leftrightarrow$E 1, 8
10	$\quad\quad \bot$	$\neg$E 7, 9
11	$\quad P$	RAA 8–10
12	$P \leftrightarrow Q$	$\leftrightarrow$I 2–6, 7–11

RAA enters at lines 6 and 11: each inner goal is a letter, and the premise produces only negations. In the converse, $P \leftrightarrow Q \vdash \neg P \leftrightarrow \neg Q$ (Example (Negating both sides of an iff)), the inner goals are negations, $\neg$I builds them, and no RAA is used.

Exercise. B3: The redundant rule

Show that $\bot$E is derivable from RAA and repetition: given a line $\bot$, derive an arbitrary statement P in one box, without $\bot$E.

Solution. Open an RAA box with $\neg P$ and repeat the $\bot$ inside it:

1	$\bot$	
2	$\neg P$	asm, for RAA
3	$\bot$	R 1
4	P	RAA 2–3

The repetition at line 3 is needed because RAA asks for $\bot$ as a line of the box itself. So every use of $\bot$E can be replaced by these three lines; the book keeps the rule for the reason given in the remark after Theorem 2.5.7.

Exercise. B4: Part ② on an example

Here is a proof by RAA about a real number x.

> Suppose $x^2 \leq x$. Suppose, for a contradiction, that $x \leq 1$ fails, that is, $x > 1$. Then $x > 0$, and multiplying $x > 1$ by x gives $x^2 > x$, which contradicts $x^2 \leq x$. Hence $x \leq 1$.

Let P be the statement "$x \leq 1$".

(a) Which sentences of the argument form the RAA box, and what are its assumption and its last line?

(b) Apply part ② of the proof of Theorem 2.5.7 to this use of RAA. Write the statement that the new line k asserts, in symbols and in words.

(c) Which rule justifies the new line k, and which rule the line after it?

(d) Which lines of the original argument are unchanged by the replacement?

Solution.

(a) The box opens with "$x \leq 1$ fails", that is $\neg P$, and runs through the sentence ending "which contradicts $x^2 \leq x$"; its last line is $\bot$, the clash between $x^2 > x$ and the hypothesis $x^2 \leq x$. The last sentence, "Hence $x \leq 1$", is the RAA step.

(b) The new line k is $\neg\neg P$, that is $\neg\neg(x \leq 1)$: "it is not the case that $x \leq 1$ fails". The line after it is P, "$x \leq 1$".

(c) Line k is $\neg$I, closing the box that opened with $\neg P$; the next line is $\neg\neg$E, citing line k.

(d) Every line of the box is unchanged: the assumption $\neg P$ ("$x \leq 1$ fails", that is $x > 1$), the steps $x > 0$ and $x^2 > x$, and the clash. Only the closing step becomes two lines. For real numbers "not not $x \leq 1$" and "$x \leq 1$" mean the same, but for the rules they are different statements, joined by $\neg\neg$E.

2.5.11.3 C. Going deeper

Exercise. C1: Excluded middle does the work of reductio directly

Suppose that, instead of RAA, we had LEM as a rule: $\varphi \vee \neg\varphi$ may be written on any line, for any statement φ.

(a) Given a box that opens with $\neg\varphi$ and reaches $\bot$, derive φ using LEM, $\vee$E, $\bot$E, and the other primitive rules, but neither RAA nor $\neg\neg$E. Write the case $\neg\varphi$ as the given box, extended.

(b) Which two parts of the proof of Theorem 2.5.7 does your fragment shortcut, and which lines of their composite does it save?

Solution (Hint 1). Write $\varphi \vee \neg\varphi$ just before the given box. What must each of the two case boxes end with?

Solution (Hint 2). The first case box is a single line.

Solution.

(a) Write excluded middle for φ just before the given box, and use the box as the second case of $\vee$E, extended by one line:

h	$\varphi \vee \neg\varphi$	LEM
$h+1$	φ	asm, for $\vee$E
i	$\neg\varphi$	asm, for $\vee$E
	$\vdots$	
j	$\bot$	
$j+1$	φ	$\bot$E j
k	φ	$\vee$E h, $h+1$–$h+1$, i–$j+1$

Here $i = h+2$. The first case is a one-line box that already ends with the goal. The second case is the given box with one more line, $\bot$E on its $\bot$; its lines are unchanged and cite what they cited before, since the box has not moved into or out of any other box. No RAA and no $\neg\neg$E is used.

(b) It shortcuts ② followed by ③. Part ② closes the box by $\neg$I, giving $\neg\neg\varphi$, then applies $\neg\neg$E; part ③ replaces that $\neg\neg$E by a split on $\varphi \vee \neg\varphi$ whose $\neg\varphi$ case is a new box that clashes with the line $\neg\neg\varphi$. The fragment in (a) uses the given box itself as the $\neg\varphi$ case, so it never writes $\neg\neg\varphi$. Counted exactly, it saves three lines: the line $\neg\neg\varphi$ of ②, and in ③ the assumption $\neg\varphi$ at line $k+2$ and the $\neg$E line $k+3$. The $\bot$E line $k+4$ of ③ survives, moved to the end of the given box.

Exercise. C2: Find the error

A student derives $\neg(\neg P \wedge \neg Q) \vdash P \vee Q$ as follows.

1	$\neg(\neg P \wedge \neg Q)$	
2	$P \vee \neg P$	LEM
3	$\quad P$	asm, for $\vee$E
4	$\quad P \vee Q$	$\vee$I 3
5	$\quad \neg P$	asm, for $\vee$E
6	$\quad \bot$	$\neg$E 3, 5
7	$\quad Q$	$\bot$E 6
8	$\quad P \vee Q$	$\vee$I 7
9	$P \vee Q$	$\vee$E 2, 3–4, 5–8

(a) Is the sequent $\neg(\neg P \wedge \neg Q) \vdash P \vee Q$ derivable?

(b) Does the student's derivation derive it? Name the first line that fails and the rule or clause it breaks.

(c) Show that the illegal step, if it were allowed, would prove too much: use the same move to "derive" $P \vee Q \vdash P$, and say why that sequent must not be derivable.

(d) Give a correct derivation.

Solution.

(a) Yes: part (d) below gives a derivation. The conclusion is true, and the argument is still wrong.

(b) No. Line 6 fails: it cites line 3, the assumption of the first case box, lines 3–4, which closed before line 5. By the scope rule of Definition 1.4.2, line 3 is not available at line 6. The form of $\neg$E is right; what is broken is availability. In words, the student used the assumption of one case inside the other case, where it has not been assumed.

(c) The same move derives $P \vee Q \vdash P$: split on the premise by $\vee$E, with a first box that opens with P and ends there, and a second box that opens with Q and "repeats" P from the first. That sequent must not be derivable: with P false and Q true the premise is true and the conclusion false, and Chapter 4 proves that no derivation leads from true premises to a false conclusion (Theorem 4.6.2). So citing a line of a sibling box is illegal in general, not just unlucky here.

(d) Keep lines 1–5, and in the case $\neg P$ we are stuck at the goal Q, so we argue for it by RAA, using the premise:

1	$\neg(\neg P \land \neg Q)$	
2	$P \lor \neg P$	LEM
3	$\quad P$	asm, for $\lor$E
4	$\quad P \lor Q$	$\lor$I 3
5	$\quad \neg P$	asm, for $\lor$E
6	$\quad\quad \neg Q$	asm, for RAA
7	$\quad\quad \neg P \land \neg Q$	$\land$I 5, 6
8	$\quad\quad \bot$	$\neg$E 7, 1
9	$\quad Q$	RAA 6–8
10	$\quad P \lor Q$	$\lor$I 9
11	$P \lor Q$	$\lor$E 2, 3–4, 5–10

Line 7 cites line 5, the assumption of the enclosing case box, which is still open: the legal version of what the student wanted.

Exercise. C3: Exit check: a classical disjunction

Derive $\vdash P \lor (P \to Q)$ using the **primitive** rules only (those of Chapter 1, $\neg$E, $\bot$E, $\neg$I, $\leftrightarrow$I, $\leftrightarrow$E, and RAA), with no derived rule, so neither LEM nor $\neg\neg$E as a single line. Name the rule on every line, and say which line is classical.

Solution. The search. The goal is a disjunction, so we postpone it; there are no premises, so no premise-directed move applies; and neither side is derivable on its own (P may be false, and $P \to Q$ is false when P is true and Q false). We are stuck, so we assume the exact negation of the goal, $\neg(P \lor (P \to Q))$, and hunt for a clash. The assumption can clash only with $P \lor (P \to Q)$, which $\lor$I builds from either side. The side $P \to Q$ is an implication, so we open a box on P with the goal Q. Inside it, $\lor$I gives $P \lor (P \to Q)$ from P, which clashes with the assumption, and $\bot$E writes Q. So $P \to Q$ holds, $\lor$I gives the goal, and that is the second clash.

1	$\quad \neg(P \lor (P \to Q))$	asm, for RAA
2	$\quad\quad P$	asm, for $\to$I
3	$\quad\quad P \lor (P \to Q)$	$\lor$I 2
4	$\quad\quad \bot$	$\neg$E 3, 1
5	$\quad\quad Q$	$\bot$E 4
6	$\quad P \to Q$	$\to$I 2–5
7	$\quad P \lor (P \to Q)$	$\lor$I 6
8	$\quad \bot$	$\neg$E 7, 1
9	$P \lor (P \to Q)$	RAA 1–8

Lines 1 and 2 are assumptions, of the RAA box and of the $\to$I box; every other line names its rule on the right. The classical line is line 9; every other rule used is a rule for a connective. The shape is that of Proposition 2.5.3: there the inner box closed by $\neg$I to give $\neg P$; here it ends with $\bot$E and closes by $\to$I to give $P \to Q$.

A second correct route first derives $P \vee \neg P$ by the eight lines of Proposition 2.5.3, written out, and then splits on it: in the case P, $\vee$I gives the goal; in the case $\neg P$, a box on P clashes with $\neg P$, $\bot$E writes Q, $\to$I gives $P \to Q$, and $\vee$I the goal. Its classical line is the RAA inside the written-out excluded middle.

Marking scheme. (10 points)

- 2: an RAA box opened with $\neg(P \vee (P \to Q))$, the exact negation of the goal.
- 3: an inner box on P that reaches $\bot$ by $\vee$I and $\neg$E against the outer assumption, and then Q by $\bot$E.
- 2: $P \to Q$ by $\to$I, then $P \vee (P \to Q)$ by $\vee$I, and the second clash.
- 2: RAA closes the outer box, and it is named as the classical step.
- 1: every citation is available.

The second route earns the same points, step for step: 2 for the written-out excluded middle opened correctly (its RAA box on $\neg(P \vee \neg P)$); 3 for the box on P in the case $\neg P$, closed by $\bot$E; 2 for $\to$I, the two $\vee$I lines, and the $\vee$E; 2 for naming the RAA inside excluded middle as the classical step; 1 for availability. Fatal errors (0 for the step): opening the outer box with anything but $\neg(P \vee (P \to Q))$; closing the outer box by $\neg$I (that gives $\neg\neg(P \vee (P \to Q))$); a line justified by $\neg\neg$E or LEM taken as a rule, which the item excludes.